

A Gm/Id based methodology for designing common source amplifier

Sampatrao L. Pinjare, Nithya.G, Veda S Nagaraja
Department of ECE
Nitte Meenakshi Institute of Technology
Bangalore,India
slpinjare@gmail.com, g.nithyaa.0706@gmail.com,
veda.nmit@gmail.com

Sthuthi A
Center for Nano Materials and MEMS
Nitte Meenakshi Institute of Technology
Bangalore,India
stuthiks@gmail.com

Abstract— Device miniaturization is of great concern over past two decades which leads to further transistor size shrinking and improvement in their performance. As the transistor size is reduced there is a need to include short channel effects. These effects are not taken into consideration by conventional equation based amplifier design methodology. An alternative approach is to use graphical based Gm/Id technique. This methodology investigates the relationship between transconductance by drain current (Gm/Id) versus normalized drain current [$I_d/(W/L)$] which is strongly related to the circuit performance. This result indicates the region of operation and also provides tool to calculate device dimensions by unified synthesis methodology in all regions of operation of a MOS transistor. In this paper, a common source amplifier with active load has been designed using Gm/Id technique to model the transistor size and meet the given design parameters such as GBW, gain and power consumption. Supply voltage of 1.8V is used here and a comparative analysis of these performance parameters is done for various technologies (180nm, 90nm, 45nm) using cadence virtuoso tool.

Keywords- Common Source Amplifier, MOSFET, Transconductance, Cadence-Virtuoso.

I. INTRODUCTION

As per the prediction of Moore's law, the number of transistors on chip has increased exponentially over past few decades while it's required to maintain low power consumption, high speed and cost reduction. In order to meet these requirements transistor sizes are reduced to a greater extent in nanometer range (current semiconductor manufacturing processes utilizes 22nm technology) for which short channel effects becomes predominant [1].

MOS modeling for transistor sizing and biasing can be done by conventional equation based technique, but it's restricted to long-channel devices. An alternative approach which provides modeling of transistors including short channel effects is known as Gm/Id technique, which is based on characterization of the transistors.

Although the trend towards miniaturization has helped digital circuitry to be densely packed with transistors, the analog counterpart had to pay the price with decreasing supply voltage headroom, reduced dynamic range, lower gain, and similar other attributes [2]. As a consequence the

conventional long-channel equations employed in the analog design no longer produce desired results.

In this work, a common source amplifier with active load is designed using Gm/Id technique. In section II Gm/Id technique is briefly illustrated followed by section III which describes design methodology of a common source amplifier using Gm/Id technique. Section IV provides the schematic and simulation results of the same.

II. GM/ID TECHNIQUE

Gm/Id method links design variables (Gm, f_t , I_d) to performance parameter (Bandwidth, power) and also employs graphs to accurately size the transistors. A methodology where transistor dimensions are obtained by fixing currents and performing complete characterization on MOS transistors to satisfy the design specifications like the gain-Bandwidth, low power, area from the derived Gm/Id ratio [3].

In long channel design, Overdrive voltage is the key parameter which determines the region of operation utilizing different set of equations where as in Gm/Id technique graphically characterize the device based on the following figure of merit.

1. Trans conductance efficiency (Gm/Id)
2. Transit frequency ($f_t = g_m/cgg$)
3. Intrinsic gain (gm/gds).

One can generate the plots with respect to these figures of merits for NMOS and PMOS characterization using cadence virtuoso tool.

III. DESIGN METHODOLOGY

In this section, a design of common source amplifier with active load is illustrated by utilizing the graphs generated from Gm/Id technique. The design specifications for this amplifier are: Unity gain bandwidth (GBW): 100MHz, slew rate (SR): 50 V/ μ s, load capacitance (CL): 1pF. The length is varied for 3 different technologies $L = 180\text{nm}$, 90nm , 45nm . The design parameters are dimensions and bias voltage of MOS Transistor. A common source amplifier with active load is shown in figure 1. Here the driving MOSFET M1 converts the variation in the input (Gate-

Source) voltage into a small signal drain current (I_d) which passes through the equivalent impedance of the circuit ($r_{o1}||r_{o2}$) and generates the amplified output voltage (1). The gain A of the CSA is given as

$$A_v = -g_{m1} (r_{o1}||r_{o2}) \quad (1)$$

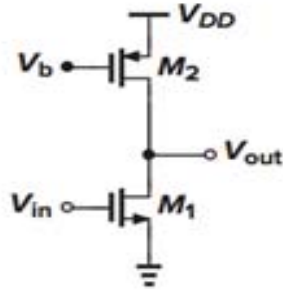


Figure 1. A circuit schematic of Common source Amplifier with active load.

The design methodology is explained here step by step for nmos based Common source amplifier with pmos load. The design is completed in following three steps.

- Design of nmos dimension for CSA using G_m/I_d technique.
- Determine dimensions of pmos load transistor.
- Determine the bias voltage for nmos/pmos transistors.

A. Finding the dimension (W) of nmos Transistor

We first model the nmos transistor as shown in figure 2 and obtain G_m/I_d Vs I_d/W , f_t vs G_m/I_d , g_m/g_{ds} vs g_m/I_d plots for $L = 180$ nm by characterizing the nmos and pmos.

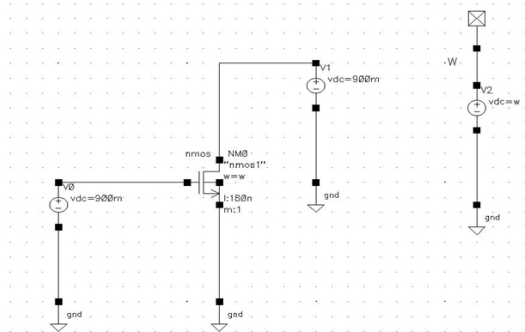


Figure 2. A circuit schematic for NMOS Characterization.

The G_m/I_d can be expressed as the ratio of the gain Bandwidth product (GBW) and the Slew rate (SR) as follows.

$$GBW = G_m/CL \quad (2)$$

$$SR = I_d/CL \quad (3)$$

from equation (2) and (3), one can obtain

$$G_m/I_d = GBW/SR = 2 \cdot \pi \cdot 100\text{MHz} \cdot 1\text{pF} / 50(\text{V}/\mu\text{s})$$

$$G_m/I_d = 12.6 \quad (4)$$

Figure 3 show a plot of I_d/W vs G_m/I_d . Using G_m/I_d value of 12.6 as obtained using equation (4), the required I_d/W is determined as 17.3. The required current is determined using Slew rate to be $50\mu\text{A}$. Then we determine W from the obtained I_d/W value, and the given current requirement as shown in equation (5).

$$W = I_d / (I_d/W) = 50\mu / 17.3 = 3\mu\text{m} \quad (5)$$

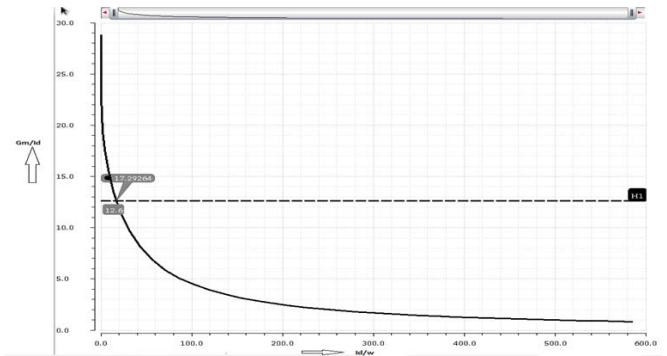


Figure 3. I_d/W vs G_m/I_d

The transit frequency (f_t) vs G_m/I_d plot is shown in figure 4. It provides useful information about the maximum speed of the device for the designed size and specified current. In order to optimize the efficiency with the speed of the circuit one needs a parameter such as transit frequency f_t to characterize the speed of the device [4]. The transit frequency can be determined for the given G_m/I_d using figure 4.

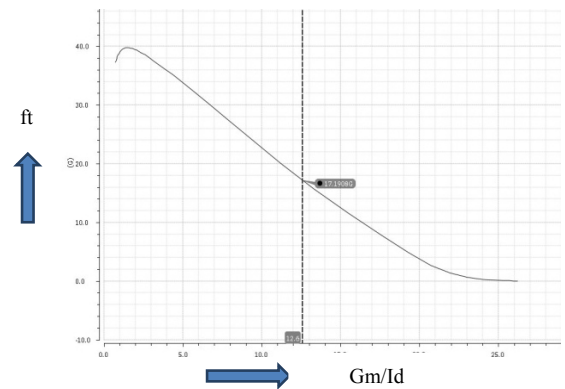


Figure 4. Transit frequency f_t vs G_m/I_d

B. Determine the dimension of pmos transistor

The mobility of nmos is 2 to 3 times greater than that of pmos, hence. Hence the width of pmos is considered to be thrice that of nmos for the same drive current [5].

C. Finding bias voltage for nmos/pmos Transistor

The bias voltage for nmos /pmos transistor is obtained by I_d vs V_{DS} graph. Here we can assume maximum dynamic range. The output voltage at operating point is assumed to be at midbias voltage (0.9 Volts). The gate bias voltage is determined from the nmos /pmos characteristics for the desired drain current (50 μ A). The schematic of circuit used for characterization of nmos is shown in figure 5. From the I_d vs V_{GS} plot for $V_{DS} = 0.9$ V, the voltage V_{GS} is determined for the desired I_d of 50 μ A. The nmos bias voltage is found to be 620mv and pmos bias voltage is 1.15V. This satisfies the mid biasing condition as shown in figure 6 I_d Vs V_{DS} plots for different V_{GS} .

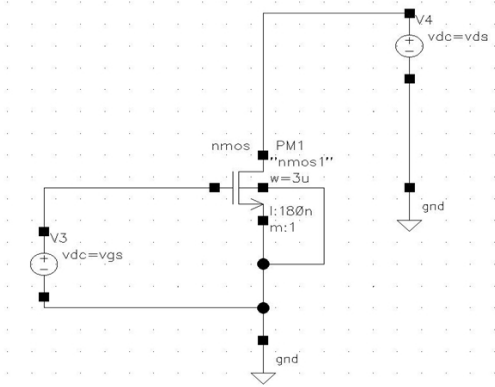


Figure 5. A circuit schematic for determination of nmos bias voltage

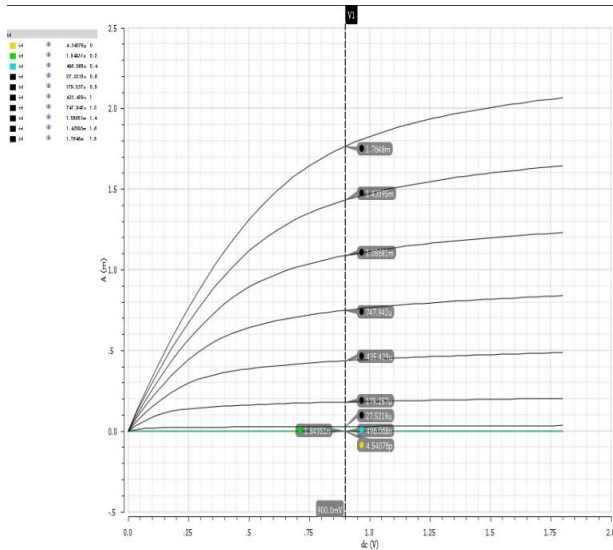


Figure 6. nmos I_d vs V_{DS} graph

IV. SCHEMATIC AND SIMULATION RESULTS

From the specifications, the device dimensions and the bias voltages for the CSA obtained are tabulated in table I.

The schematic of CSA using these dimensions is shown in figure 7. This schematic has been used for DC, transient and AC analysis of CSA to examine whether the desired specifications are met.

TABLE I. Designed device parameters for nmos and pmos

Device	Aspect ratio	Bias Voltage
nmos	16.67	620mV
pmos	50	1.15V

This schematic has been used for DC, transient and AC analysis of CSA to examine whether the desired specifications are met.

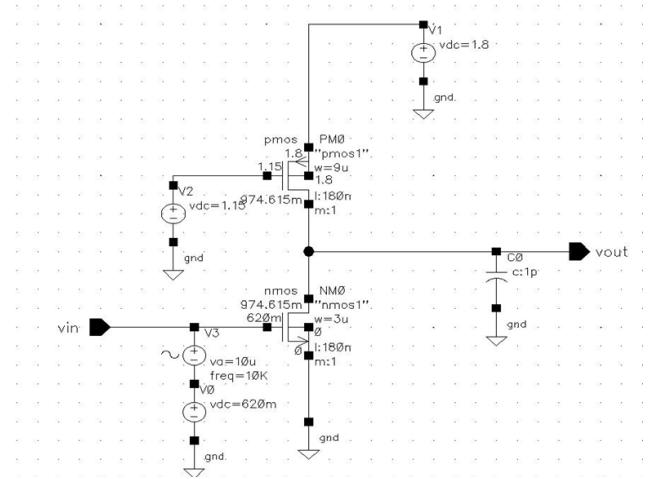


Figure 7. A circuit schematic of CSA with active load

A. DC Analysis

The dc analysis is carried out for input voltage V_{in} ranging from 0 to 1.8V and graph is plotted for V_{in} vs V_{out} as shown in figure 8. The various design parameters are theoretically calculated and compared with the obtained simulation results where both the results closely matches each other are show in table II.

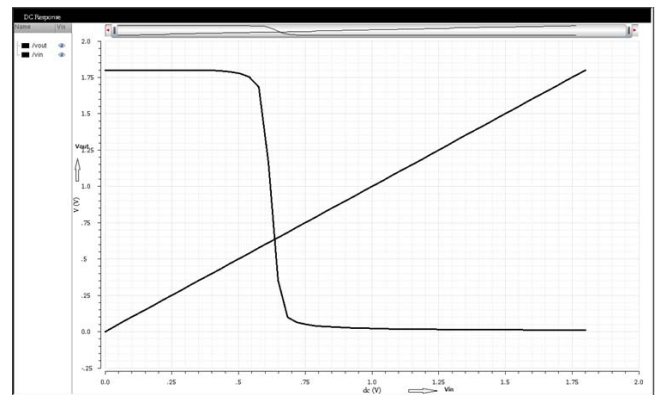


Figure 8. DC Analysis: Transfer characteristics of CSA

TABLE II. Comparison of design parameters for nmos/pmos

Design Variables	nmos		pmos
	Designed	Simulated	
gm	630 μ S	643 μ S	580 μ S
Id	50 μ A	51.19 μ A	-51.126 μ A
Gm/Id	12.6	12.56	11.34
VDS	900mV	974mV	-825.385mV

B. Transient Analysis

Transient analysis is carried out to see the input output behavior of CSA with a sinusoidal input source of 10 μ V, 10 kHz. An amplified output signal of gain 27dB was observed as shown in figure 9.

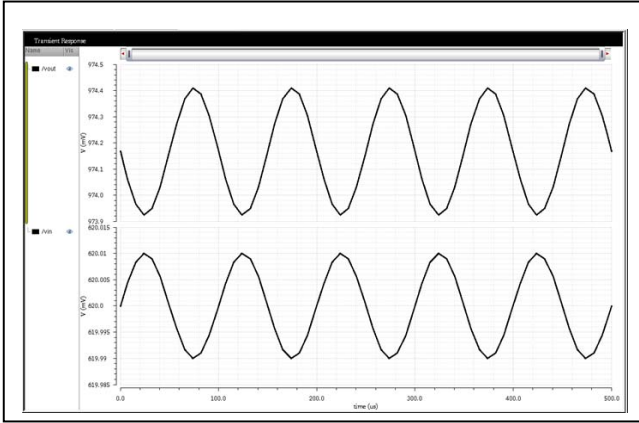


Figure 9. Transient analysis: Input and output voltage of CSA

C. AC Analysis

By carrying out ac analysis the frequency response graph for the circuit is generated and the unit gain Bandwidth (GBW) is noted to be 101.3MHz as shown in the figure 10.

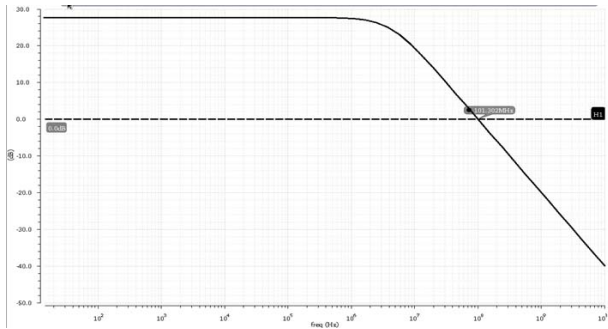


Figure 10. AC analysis: Frequency response of CSA.

Similar analysis were carried out at 90nm, 45nm technology using Gm/Id technique and it is observed that the methodology satisfies the design constraint for all the technologies(180nm, 90nm, 45nm) as tabulated in Table III. In all the technology GBW obtained is nearly 100MHz which satisfies the desired design specifications.

TABLE III. Comparison table for device parameters at various technologies

	180nm	90nm	45nm
NMOS Dimension(W)	3u	1.5u	2.5u
PMOS Dimension(W)	9u	4.5u	7.5u
GBW	101.3MHz	102.348MHz	100.8MHz
Gain	27.7dB	12.05dB	16.34dB
Power consumed	92uW	50uW	51uW

V. CONCLUSION

In this work a comparative study on different technologies (180nm, 90nm, 45nm) using Gm/Id technique has been carried out to analyze the behavior of CSA for a given specification. It is observed that the specified GBW has been achieved at all technologies satisfying the requirement. Thus Gm/Id technique allows us to design the analog circuit in nanometer scale devices; hence miniaturization of transistor and meeting the required performance parameters can be achieved parallelly.

REFERENCES

- [1] Behzad Razavi, "Design of analog CMOS integrated circuits", McGraw-Hill Education, 2001.
- [2] Arun J. Rao, "Analog Front-End Design Using the gm/ID Method for a Pulse-Based Plasma Impedance Probe System", Utah State University, All graduate theses and dissertation, June 2010.
- [3] F. Silveira, D. Flandre, and P. G. A. Jespers "A Gm/ID methodology for the design of CMOS Analog circuits and its application to the synthesis of a Silicon-on- Insulator Micropower OTA, IEEE journal of solid-state circuits, vol. 31, no. 9, September 1996.
- [4] Daniel Foty, David Binkley, and Matthias Bucher "Starting Over: gm/Id-Based MOSFET Modeling as a Basis for Modernized Analog Design Methodologies", Research gate, May 2014.
- [5] P. E. Allen and D. R. Holberg, "CMOS Analog Circuit Design", NewYork Holt Rinehart and Winston, 1987.